

CFD HW #3 20244109 기계공학과 권혁수

Consider the ordinary differential equation

$$\frac{d^2 T}{dx^2} + S(x) = 0$$

where

$$S(x) = \begin{cases} 1 & \text{at } 0 \leq x \leq 1/2 \\ 0 & \text{at } 1/2 \leq x \leq 1 \end{cases}$$

subject to the boundary conditions

$$T(0) = 0 \quad T(1) = 50$$

(1) Obtain the solution to the same differential equation numerically by using 40 control

volumes of equal size $\Delta x = 0.025$

(2) Compare them graphically with the analytical solution.

(1) Control Volume Method

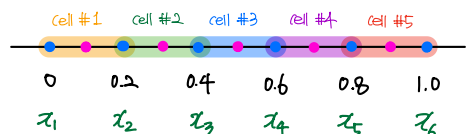
Before generating 40 control volumes, I'll first manually create 5 control volumes and obtain the solution. Based on this, I'll then explain how to write code that can handle 40 control volumes.

Step 1. Discretize the domain

$$\Delta x = \frac{1}{5} = 0.2$$

$$x_2, x_3 \rightarrow S = 1$$

$$x_4, x_5 \rightarrow S = 0$$



$$S_i = \begin{cases} 1 & (i = 1, 2, 3) \\ 0 & (i = 4, 5, 6) \end{cases} \rightarrow \text{We can apply stepwise profile for the source term.} \\ (\bar{S} = S_i)$$

Step 2. Integrate the differential equation

$$\frac{d^2 T}{dx^2} + S(x) = 0 \xrightarrow{\text{integrate}} \frac{dT}{dx} \Big|_{x_{i+1/2}} - \frac{dT}{dx} \Big|_{x_{i-1/2}} + \int_{x_{i-1/2}}^{x_{i+1/2}} S(x) dx = 0$$

Step 3. Select piecewise profile

Let's assume piecewise linear profile for $\frac{d^2 T}{dx^2}$.

$$\frac{T_{i+1} - T_i}{\Delta x} - \frac{T_i - T_{i-1}}{\Delta x} + \bar{S} \Delta x = 0 \rightarrow \left(2 \frac{1}{\Delta x} \right) T_i = \left(\frac{1}{\Delta x} \right) T_{i-1} + \left(\frac{1}{\Delta x} \right) T_{i+1} + \frac{\bar{S} \Delta x}{b} \\ \rightarrow \underline{a_i T_i = a_{i-1} T_{i-1} + a_{i+1} T_{i+1} + b \quad (a_i, a_{i-1}, a_{i+1} > 0)} \\ \text{or } T_{i+1} - 2T_i + T_{i-1} = -S_i (\Delta x)^2$$

* The Four Basic Rules.

Rule 1: Consistency at control volume faces. $\rightarrow$ It is always satisfied since I used piecewise linear profile and the thermal conductivity is uniform throughout the entire domain.

Rule 2: Positive coefficients $\rightarrow \left(2 \frac{1}{\Delta x} \right) T_i = \left(\frac{1}{\Delta x} \right) T_{i-1} + \left(\frac{1}{\Delta x} \right) T_{i+1} + S_i \Delta x \rightarrow$ All coefficients are positive. It satisfies the rule #2.

Rule 3: Negative-slope linearization of the source term $\rightarrow \bar{S} = S_i + \underline{0} \cdot T_i \rightarrow$ The coefficient of T_i is equal to zero. So, it satisfies the rule #3.

Rule 4: Sum of the neighbor coefficients $\rightarrow 2 \frac{1}{\Delta x} (T_i + \underline{C}) = \frac{1}{\Delta x} (T_{i-1} + \underline{C}) + \frac{1}{\Delta x} (T_{i+1} + \underline{C}) + S_i \Delta x$

The rule #4 is also satisfied, because $T + C$, where C is a constant, is also a solution to the above algebraic equation.

$\rightarrow$ By the four basic rules, the solution will converge.

i) node #2

$$S_2 = 1 \longrightarrow T_3 - 2T_2 + T_1 = -1 \cdot (0.2)^2 = -0.04$$

$$\text{B.C. : } T_1 = 0 \longrightarrow \underline{-2T_2 + T_3 = -0.04}$$

iii) node #4

$$S_4 = 0 \longrightarrow \underline{T_3 - 2T_4 + T_5 = 0}$$

ii) node #3

$$S_3 = 1 \longrightarrow T_4 - 2T_3 + T_2 = -1 \cdot (0.2)^2 = -0.04$$

$$\underline{T_2 - 2T_3 + T_4 = -0.04}$$

iv) node #5

$$S_5 = 0 \longrightarrow T_4 - 2T_5 + T_6 = 0$$

$$\text{B.C. : } T_6 = 50 \longrightarrow \underline{T_4 - 2T_5 = -50}$$

Step 4. Solve the discretization equations

$$\left\{ \begin{array}{ll} -2T_2 + T_3 = -0.04 & \text{--- ①} \\ T_2 - 2T_3 + T_4 = -0.04 & \text{--- ②} \\ T_3 - 2T_4 + T_5 = 0 & \text{--- ③} \\ T_4 - 2T_5 = -50 & \text{--- ④} \end{array} \right. \quad \begin{array}{l} \text{①} + 2 \times \text{②} \longrightarrow -3T_3 + 2T_4 = -0.12 \\ 2 \times \text{③} + \text{④} \longrightarrow 2T_3 - 3T_4 = -50 \end{array} \} \longrightarrow T_3 = 20.072, T_4 = 30.048$$

$$\downarrow$$

$$T_2 = 10.056, T_5 = 40.024$$

$$(\text{B.C.s : } T_1 = 0, T_6 = 50)$$

$$\therefore T_2 = 10.056, T_3 = 20.072, T_4 = 30.048, T_5 = 40.024$$

Now, we achieved the solution using 5 control volumes.

In the above, the solution was obtained using a system of equations, but we can also consider it in matrix form as follows.

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \end{bmatrix} \longrightarrow \begin{bmatrix} -2 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} T_2 \\ T_3 \\ T_4 \\ T_5 \end{bmatrix} = \begin{bmatrix} b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}$$

$\hookrightarrow -T_6 = -50$

Since T_1 and T_6 are known from the boundary conditions, the left-hand algebraic equations are reduced to

the matrix on the right, which considers only the internal nodes. Here, we can observe that all the diagonal elements are -2 , and the elements on both the upper and lower diagonals are 1 . Additionally, the last value of the $\tilde{b}$ vector is assigned as the negative of T_6 . Now, using these characteristics, we will proceed to compute the numerical solution using 40 control volumes. Let's move on to the next step.

From now on, we are going to obtain the solution by using 40 control volumes.

First, we construct a matrix with 39 rows and columns, corresponding to the number of internal nodes.

Then, we formulate the algebraic equations by incorporating the characteristics identified above and solve this linear system. The following is the pseudocode for solving the algebraic equations obtained through the FVM.

< pseudo algorithm coding >

* INPUT

$$L = 1$$

$$T_0 = 0$$

$$T_{40} = 50$$

$$S(x) = \{ 1 \text{ if } x \leq 0.5, 0 \text{ if } x > 0.5 \}$$

$$\text{Num_CV} = 40$$

* OUTPUT

$$T_{\text{num}}(x)$$

* STEP 1. Grid Generation

$$\text{Num_nodes} = \text{Num_CV} + 1$$

$$\Delta x = L / \text{Num_CV} \quad \# \Delta x = 0.025$$

for $i = 1$ to Num_nodes do

$$x[i] = \Delta x(i - 1)$$

end for



* STEP 2. Source term (stepwise profile)

for $i = 1$ to Num_nodes do

if $x[i] \leq 0.5$ then

$$S[i] = 1$$

else

$$S[i] = 0$$

end if

end for

* STEP 3. Finite Volume Formulation

$$\begin{bmatrix} -2 & 1 & 0 & \dots & 0 \\ 1 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & \vdots \\ \vdots & & 0 & \ddots & 1 & 0 \\ 0 & 0 & \dots & 1 & -2 & 1 \\ 0 & 0 & \dots & 0 & 1 & -2 \end{bmatrix} \cdot \begin{bmatrix} T_2 \\ T_3 \\ T_4 \\ \vdots \\ T_{39} \end{bmatrix} = \begin{bmatrix} B_2 \\ B_3 \\ B_4 \\ \vdots \\ B_{38} \\ B_{39} \end{bmatrix}$$

$\hookrightarrow -T_{40}$

$$\text{Num_interior} = \text{Num_nodes} - 2$$

$$\# \text{Num_interior} = 39$$

$$A = \text{zeros}(\text{Num_interior}, \text{Num_interior})$$

$$b = \text{zeros}(\text{Num_interior}, 1)$$

for $i = 1$ to Num_interior do

$$\# \text{Num_interior} = 39$$

$$A[i][i] = -2$$

if $i > 1$ then

$$A[i][i-1] = 1$$

end if

if $i < \text{Num_interior}$ then

$$A[i][i+1] = 1$$

end if

$$B[i] = -S[i+1] * (\Delta x)^2$$

end for

* STEP 4. Boundary conditions

$$b[\text{Num_interior}] = -T_{40}$$

* STEP 5. Solve the linear system

$$T_{\text{internal}} = \text{solve}(A, b)$$

* STEP 6. Total T formulation

$$T_{\text{num}}[1] = T_0$$

for $i = 2$ to $\text{Num_node} - 1$ do

$$\# \text{Num_node} = 41$$

$$T_{\text{num}}[i] = T_{\text{internal}}[i - 1]$$

end for

$$T_{\text{num}}[\text{Num_node}] = T_{40}$$

< MATLAB >

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1 clear; close all; clc;
2
3 %% STEP 1. Grid Generation
4 N = 41; % Total number of nodes 40 + 1 = 41
5 L = 1; % Domain Length
6 dx = L / (N - 1); % dx = 1/40 = 0.025
7 x = linspace(0, L, N)'; % Location of Nodes
8
9 %% STEP 2. Source vector generation using Stepwise profile
10 S = ones(N,0);
11 S(x > 0.5) = 0;
12 S(x <= 0.5) = 1;
13
14 %% STEP 3. Finite Volume Formulation
15
16 %  $T(i-1) - 2T(i) + T(i+1) = -S_i * (dx)^2$ 
17
18 Num_int_nodes = N - 2; % 41 - 2 = 39
19
20 % Initialization (Ax=b)
21 A = zeros(Num_int_nodes, Num_int_nodes);
22 b = zeros(Num_int_nodes, 1);
23
24 % Matrix A formulation
25 for i = 1:Num_int_nodes
26     A(i,i) = -2; % diagonal element
27     if i > 1
28         A(i, i-1) = 1; % lower diagonal element
29     end
30     if i < Num_int_nodes
31         A(i, i+1) = 1; % upper diagonal element
32     end
33 end
34
35 % Vector b formulation
36
37 %  $b[1] = b_2, b[2] = b_3, \dots, b[39] = -T_{40} = -50$ 
38
39 for i = 1:Num_int_nodes
40     index_change = i + 1;
41     b(i) = -S(index_change) * dx^2;
42 end
43
44 b(Num_int_nodes) = -50;
45
46 %% STEP 4. Solving Linear System
47
48 %  $A*x = b \rightarrow x = A \backslash b \rightarrow x = T_{int}$ 
49 T_int = linsolve(A, b);
50
51 % T formulation (BCs + interior)
52 T = zeros(N,1);
53 T(1) = 0; % Boundary condition
54 T(2:N-1) = T_int; % Temperature of internal nodes
55 T(N) = 50; % Boundary condition
56
57 fprintf('Temperature at each node:\n');
58 for i = 1:N
59     fprintf('T%d = %.6f\n', i, T(i));
60 end
61
62 figure;
63 plot(x, T, 'bo-', 'LineWidth', 2);
64 grid on;
65
66 xlabel('x');
67 ylabel('Temperature T(x)');
68 title('Numerical Solution using 40 control volumes');
69
70

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* Solution

```

Temperature at each node:
T1 = 0.000000
T2 = 1.259219
T3 = 2.517812
T4 = 3.775781
T5 = 5.033125
T6 = 6.289844
T7 = 7.545937
T8 = 8.801406
T9 = 10.056250
T10 = 11.310469
T11 = 12.564062
T12 = 13.817031
T13 = 15.069375
T14 = 16.321094
T15 = 17.572187
T16 = 18.822656
T17 = 20.072500
T18 = 21.321719
T19 = 22.570312
T20 = 23.818281
T21 = 25.065625
T22 = 26.312344
T23 = 27.559062
T24 = 28.805781
T25 = 30.052500
T26 = 31.299219
T27 = 32.545937
T28 = 33.792656
T29 = 35.039375
T30 = 36.286094
T31 = 37.532812
T32 = 38.779531
T33 = 40.026250
T34 = 41.272969
T35 = 42.519687
T36 = 43.766406
T37 = 45.013125
T38 = 46.259844
T39 = 47.506562
T40 = 48.753281
T41 = 50.000000

```

(2) Analytical solution

i) $0 \leq x \leq \frac{1}{2}$

$$\frac{dT^2}{dx^2} + 1 = 0 \xrightarrow{\text{integration}} \frac{dT}{dx} = -x + C_1 \xrightarrow{\text{integration}} T(x) = -\frac{1}{2}x^2 + C_1x + C_2$$

B.C : $T(0) = 0 \longrightarrow T(0) = C_2 = 0$

For continuity at $x = \frac{1}{2}$, $T(\frac{1}{2}) = -\frac{1}{8} + \frac{1}{2}C_1 \dots \textcircled{1}$

$T'(\frac{1}{2}) = -\frac{1}{2} + C_1 \dots \textcircled{2}$

ii) $\frac{1}{2} < x \leq 1$

$$\frac{dT^2}{dx^2} = 0 \xrightarrow{\text{integration}} \frac{dT}{dx} = C_3 \xrightarrow{\text{integration}} T(x) = C_3x + C_4$$

B.C : $T(1) = 50 \longrightarrow T(1) = C_3 + C_4 = 50 \dots \textcircled{3}$

For continuity at $x = \frac{1}{2}$, $T(\frac{1}{2}) = \frac{1}{2}C_3 + C_4 = \frac{1}{2}C_3 + (50 - C_3) \dots \textcircled{4}$

$T'(\frac{1}{2}) = C_3 \dots \textcircled{5}$

$\textcircled{1} = \textcircled{4} \longrightarrow -\frac{1}{8} + \frac{1}{2}C_1 = -\frac{1}{2}C_3 + 50 \longrightarrow C_1 + C_3 = 100.25$

$\textcircled{2} = \textcircled{5} \longrightarrow -\frac{1}{2} + C_1 = C_3 \longrightarrow C_1 - C_3 = 0.5$

$$\oplus \begin{cases} C_1 + C_3 = 100.25 \\ C_1 - C_3 = 0.5 \end{cases} \xrightarrow{2C_1} 2C_1 = 100.75 \longrightarrow C_1 = 50.375, C_3 = 49.875 \longrightarrow \therefore C_4 = 0.125$$

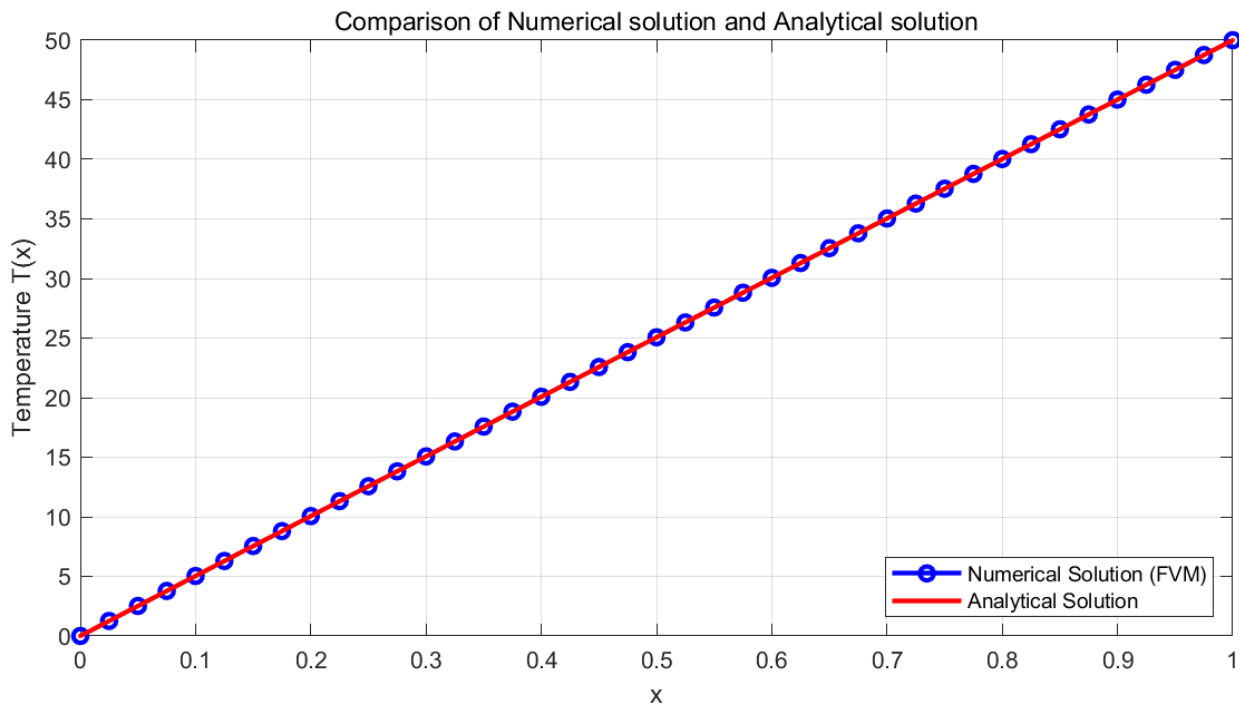
$$T(x) = \begin{cases} -\frac{1}{2}x^2 + 50.375x & (0 \leq x \leq \frac{1}{2}) \\ 49.875x + 0.125 & (\frac{1}{2} < x \leq 1) \end{cases}$$

```

71 %% Analytical Solution
72
73 T_analytic = zeros(N,1);
74 for i = 1:N
75     if x(i) <= 0.5
76         T_analytic(i) = -0.5 * x(i)^2 + 50.375 * x(i);
77     else
78         T_analytic(i) = 49.875 * x(i) + 0.125;
79     end
80 end
81
82 %% Plot and Comparison
83 figure;
84 plot(x, T, 'bo-', 'linewidth', 2); hold on;
85 plot(x, T_analytic, 'r.-', 'linewidth', 2);
86 grid on;
87
88 xlabel('x');
89 ylabel('Temperature T(x)');
90 title('Comparison of Numerical solution and Analytical solution');
91 legend('Numerical Solution (FVM)', 'Analytical Solution', Location='southeast');
92

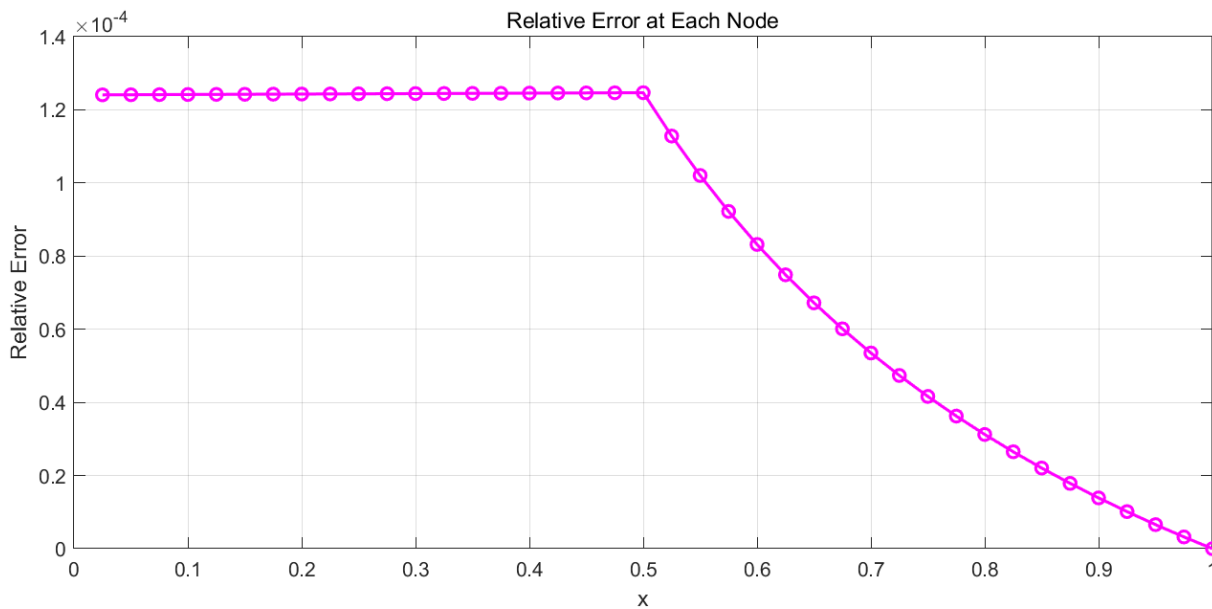
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The following plot shows both the numerical and analytical solutions for comparison.



To compare the two solutions, the RMS error and the relative error at each node were calculated.

RMS Error between numerical and analytical solution: 0.001783



As shown in the relative error plot above, the error at all nodes was less than 0.001.

Within the range where the source term has a value of 1, the relative error remains constant at approximately 0.000124 across all nodes. However, once the source term becomes zero beyond $x = 0.5$, the relative error starts to decrease gradually and approaches zero. The result shows that the numerical solution obtained using 40 control volumes has a very small error compared to the analytical solution.